Task-4p

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0.1 Deakin University

0.2 SIG731- Data Wrangling

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0.3 Task-4p

Objective

The nycflights13_weather.csv.gz dataset provides hourly meteorological data for three major airports in New York City (LGA, JFK, and EWR) for the entire year of 2013. The dataset includes various weather-related variables such as temperature, dew point, humidity, wind speed, precipitation, pressure, and visibility. The primary objective of this dataset is to analyze and understand weather patterns at these airports.

- **Study Weather Trends:** Analyze how weather conditions (temperature, wind speed, precipitation) vary over time (daily, monthly ,Hour).
- **Impact on Aviation:** Investigate how weather conditions might affect flight operations, such as delays, cancellations, or safety.
- **Compare Airports**: Compare weather conditions across the three airports (LGA, JFK, and EWR) to identify differences or similarities.
- **Data Cleaning and Preparation:** Address issues like the incorrect time_hour column and outliers in the data, which are common challenges in real-world datasets.

0.4 Importting Library Package

[18]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib.dates as mdates

#Python warning ingorning
import warnings
warnings.filterwarnings('ignore')
```

```
pd.set_option("display.notebook_repr_html", False)
```

[19]:
```
weather_df = pd.read_csv("data/nycflights13_weather.csv.gz",comment="#")
weather_df.head()
```

[19]:
```
  origin  year  month  day  hour   temp   dewp  humid  wind_dir  wind_speed  \
0    EWR  2013      1    1     0  37.04  21.92  53.97     230.0    10.35702
1    EWR  2013      1    1     1  37.04  21.92  53.97     230.0    13.80936
2    EWR  2013      1    1     2  37.94  21.92  52.09     230.0    12.65858
3    EWR  2013      1    1     3  37.94  23.00  54.51     230.0    13.80936
4    EWR  2013      1    1     4  37.94  24.08  57.04     240.0    14.96014

   wind_gust  precip  pressure  visib            time_hour
0  11.918651     0.0    1013.9   10.0  2013-01-01 01:00:00
1  15.891535     0.0    1013.0   10.0  2013-01-01 02:00:00
2  14.567241     0.0    1012.6   10.0  2013-01-01 03:00:00
3  15.891535     0.0    1012.7   10.0  2013-01-01 04:00:00
4  17.215830     0.0    1012.8   10.0  2013-01-01 05:00:00
```

Dataset

Here is the nycflights13_weather.csv.gz data file loaded It gives the hourly meteorological data for three airports in New York: LGA, JFK, and EWR for the whole year of 2013.

Columns:

- origin – weather station: LGA, JFK, or EWR,
- year, month, day, hour – time of recording,
- temp, dewp – temperature and dew point in degrees Fahrenheit,
- humid – relative humidity,
- wind_dir, wind_speed, wind_gust – wind direction (in degrees), speed and gust speed (in mph),
- precip – precipitation, in inches,
- pressure – sea level pressure in millibars,
- visib – visibility in miles,
- time_hour – date and hour (based on the year, month, day, hour fields) formatted as YYYY-mm-dd HH:MM:SS (actually, YYYY-mm-dd HH:00:00).

0.4.1 Q1. Convert all columns so that they use metric (International System of Units, SI) or derived units: temp and dewp to Celsius, precip to millimetres, visib to metres.

[20]:
```
# International System of Units, SI conversions using lambda functions
weather_df['temp'] = weather_df['temp'].apply(lambda x: (x - 32) * 5 / 9)
weather_df['dewp'] = weather_df['dewp'].apply(lambda x: (x - 32) * 5 / 9)
weather_df['precip'] = weather_df['precip'].apply(lambda x: x * 25.4)
weather_df['visib'] = weather_df['visib'].apply(lambda x: x * 1609.34)
weather_df['wind_speed'] = weather_df['wind_speed'].apply(lambda x: x * 0.44704)
weather_df['wind_gust'] = weather_df['wind_gust'].apply(lambda x: x * 0.44704)
```

```
# Convert time_hour to datetime
weather_df['time_hour'] = pd.to_datetime(weather_df['time_hour'])
weather_df
```

[20]:

	origin	year	month	day	hour	temp	dewp	humid	wind_dir	wind_speed	wind_gust	precip	pressure	visib	time_hour
0	EWR	2013	1	1	0	2.8	-5.6	53.97	230.0	4.630002	5.328114	0.0	1013.9	16093.4	2013-01-01 01:00:00
1	EWR	2013	1	1	1	2.8	-5.6	53.97	230.0	6.173336	7.104152	0.0	1013.0	16093.4	2013-01-01 02:00:00
2	EWR	2013	1	1	2	3.3	-5.6	52.09	230.0	5.658892	6.512139	0.0	1012.6	16093.4	2013-01-01 03:00:00
3	EWR	2013	1	1	3	3.3	-5.0	54.51	230.0	6.173336	7.104152	0.0	1012.7	16093.4	2013-01-01 04:00:00
4	EWR	2013	1	1	4	3.3	-4.4	57.04	240.0	6.687781	7.696165	0.0	1012.8	16093.4	2013-01-01 05:00:00
…	…	…	…	…	…	…	…	…	…	…	…	…	…	…	…
26125	LGA	2013	12	30	19	2.2	-6.7	51.78	340.0	6.173336	7.104152	0.0	1017.1	16093.4	2013-12-30 20:00:00
26126	LGA	2013	12	30	20	1.1	-8.3	49.51	330.0	7.716670	8.880190	0.0	1018.8	16093.4	2013-12-30 21:00:00
26127	LGA	2013	12	30	21	0.0	-9.4	49.19	340.0	6.687781	7.696165	0.0	1019.5	16093.4	2013-12-30 22:00:00
26128	LGA	2013	12	30	22	-0.6	-10.6	46.74	320.0	7.716670	8.880190	0.0	1019.9	16093.4	2013-12-30 23:00:00
26129	LGA	2013	12	30	23	-1.7	-11.7	46.41	330.0	8.231115	9.472203	0.0	1020.9	16093.4	2013-12-31 00:00:00

[26130 rows x 15 columns]

Here, I have convert the columns to metric SI units using lamda function formual is apply like bellow Temperature Convert from Fahrenheit to Celsius.

$C = (F - 32)\frac{9}{5}$

temp and dewp colum has converted

Precipitation (precip): Convert from inches to millimeters.

$ mm=inches×25.4$

Visibility (visib): Convert from miles to meters.

$meters = miles \times 1609.34$

Miles are converted to meters by multiplying by 1609.34 (since 1 mile = 1609.34 meters)

Wind Speed (wind_speed and wind_gust): Convert from miles per hour (mph) to meters per second (m/s).

Formula:

$m/s = mph \times 0.44704$

Miles per hour (mph) are converted to meters per second (m/s) by multiplying by 0.44704 (since 1 mph = 0.44704 m/s).

Q2. Compute daily mean wind speeds for the LGA airport (~365 total speed values, for each day separately

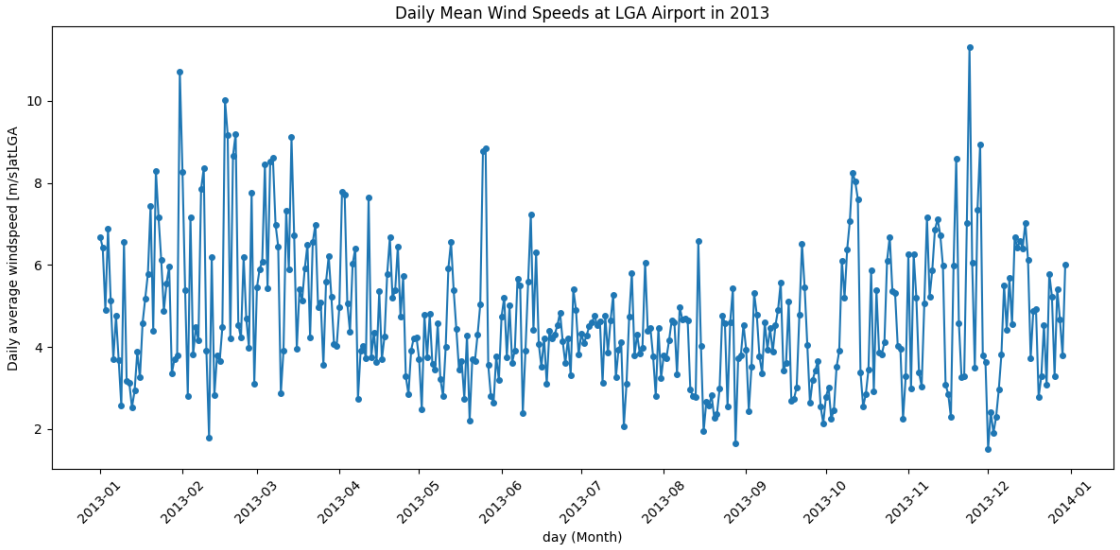
[21]:
```
# Filter for LGA of Airport
lga_weather = weather_df[weather_df['origin'] == 'LGA']

# Group by year, month, and day, and compute mean wind speed
daily_mean_wind_speed = lga_weather.groupby(['year', 'month',␣
 ↪'day'])['wind_speed'].mean().reset_index()

# Rename the mean column for clarity
daily_mean_wind_speed.rename(columns={'wind_speed': 'mean_wind_speed'},␣
 ↪inplace=True)

# Display the result
print(daily_mean_wind_speed)
```

```
     year  month  day  mean_wind_speed
0    2013      1    1         6.687781
1    2013      1    2         6.430559
2    2013      1    3         4.908660
3    2013      1    4         6.880698
4    2013      1    5         5.144447
..    …    …  …  …              …
359  2013     12   26         3.301020
360  2013     12   27         5.401669
361  2013     12   28         4.672873
362  2013     12   29         3.794030
363  2013     12   30         6.001855

[364 rows x 4 columns]
```

- Filter the data for the LGA airport.
- Group the data by year, month, and day.
- Compute the mean wind speed for each day.
- Print the Daily (each day) Mean wind sepeed

0.4.2 Q3.Present the daily mean wind speeds at LGA (~365 aforementioned data points) in a single plot, using the matplotlib.pyplot.plot function

[22]:
```
# Filter for LGA airport
lga_weather = weather_df[weather_df['origin'] == 'LGA']

# Group by year, month, and day, and compute mean wind speed
daily_mean_wind_speed = lga_weather.groupby(['year', 'month',␣
 ↪'day'])['wind_speed'].mean().reset_index()

# Create a datetime column for plotting
daily_mean_wind_speed['date'] = pd.to_datetime(daily_mean_wind_speed[['year',␣
 ↪'month', 'day']])

# Plot line grapsh
plt.figure(figsize=(12, 6))
plt.plot(daily_mean_wind_speed['date'], daily_mean_wind_speed['wind_speed'],␣
 ↪marker='o', linestyle='-', markersize=4)

# Format x-axis to show month column
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%b'))
plt.gca().xaxis.set_major_locator(mdates.MonthLocator())

# Add labels and title
plt.xlabel('day (Month)')
plt.ylabel('Daily average windspeed [m/s]atLGA')
plt.title('Daily Mean Wind Speeds at LGA Airport in 2013')

# Format x-axis to show year and month (YYYY-MM)
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m'))
plt.gca().xaxis.set_major_locator(mdates.MonthLocator())

# Rotate x-axis labels for better readability
plt.xticks(rotation=45)

# Show the plot
plt.tight_layout()
plt.show()
```



Here is step for description

- Filter for LGA airport
- Group by year, month, and day, and compute mean wind speed
- Create a datetime column for plotting
- Plot line grapsh
- Format x-axis to show month column
- Format x-axis to show year and month (YYYY-MM)
- Rotate x-axis labels for better readability
- Show the plot

0.4.3 Q4. Identify the ten windiest days at LGA (dates and the corresponding mean daily wind speeds).

[23]:
```
# Filter for LGA airport
lga_weather = weather_df[weather_df['origin'] == 'LGA']

# Group by year, month, and day, and compute mean wind speed
daily_mean_wind_speed = lga_weather.groupby(['year', 'month',␣
 ↪'day'])['wind_speed'].mean().reset_index()

# New datetime column  add
daily_mean_wind_speed['date'] = pd.to_datetime(daily_mean_wind_speed[['year',␣
 ↪'month', 'day']])

# Sort by mean wind speed in descending order
windiest_days = daily_mean_wind_speed.sort_values(by='wind_speed',␣
 ↪ascending=False)
```

```
# Format the output
windiest_days = windiest_days[['date', 'wind_speed']].reset_index(drop=True)
windiest_days['date'] = windiest_days['date'].dt.strftime('%Y-%m-%d')

# Display the result
print(windiest_days)
```

```
           date  wind_speed
0    2013-11-24   11.317783
1    2013-01-31   10.717598
2    2013-02-17   10.010236
3    2013-02-21    9.192903
4    2013-02-18    9.174264
..          …          …
359  2013-08-16    1.945943
360  2013-12-03    1.907732
361  2013-02-11    1.800556
362  2013-08-28    1.650510
363  2013-12-01    1.521899

[364 rows x 2 columns]
```

0.4.4 Q5. Compute the monthly mean wind speeds for all the three airports.

[24]:
```
# Identify the outlier Calculating InterQuartile
Q1 = weather_df['wind_speed'].quantile(0.25)
Q3 = weather_df['wind_speed'].quantile(0.75)
IQR = Q3 - Q1

# Define the outlier threshold lower bound and upper bound
lb = Q1 - 1.5 * IQR
ub = Q3 + 1.5 * IQR

# Replace the outlier with NaN
weather_df['wind_speed'] = np.where(
    (weather_df['wind_speed'] < lb) | (weather_df['wind_speed'] > ub),
    np.nan,
    weather_df['wind_speed']
)

# Group by origin, year, and month, and compute mean wind speed
monthly_mean_wind_speed = weather_df.groupby(['origin', 'year',␣
 ↪'month'])['wind_speed'].mean().reset_index()

# Result
print(monthly_mean_wind_speed)
```

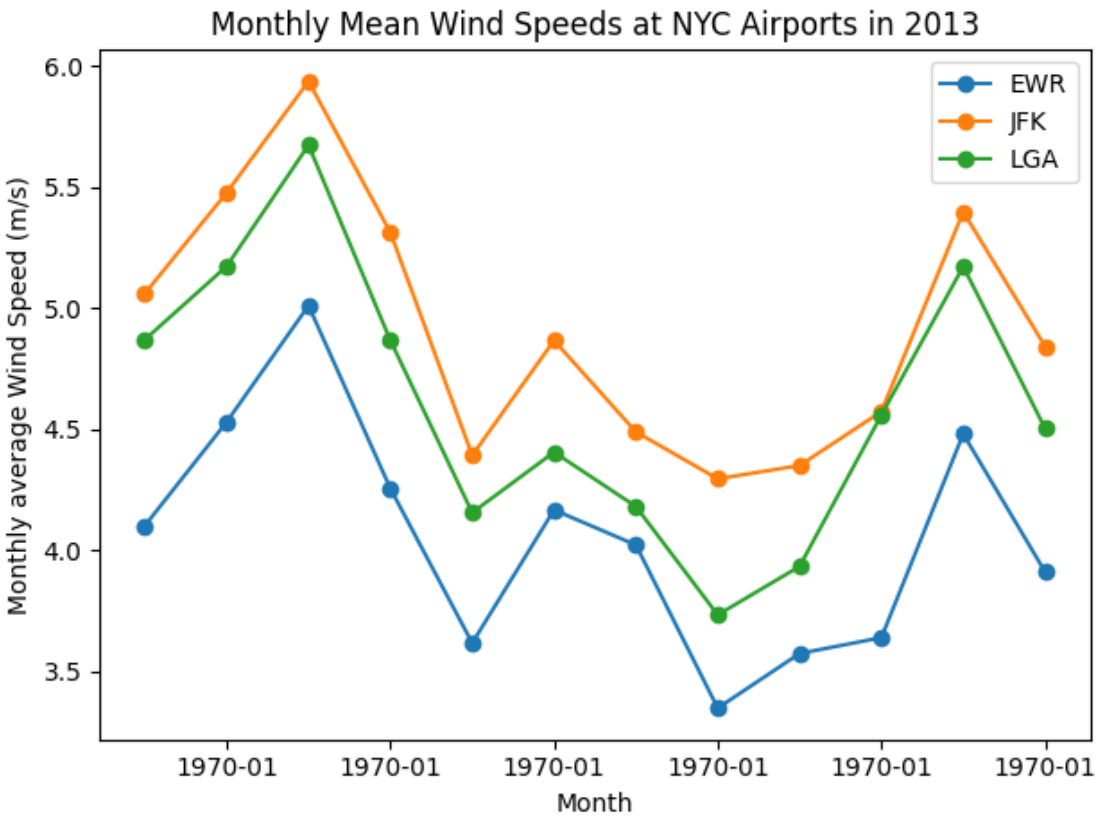
	origin	year	month	wind_speed
0	EWR	2013	1	4.102076
1	EWR	2013	2	4.530279
2	EWR	2013	3	5.010183
3	EWR	2013	4	4.255796
4	EWR	2013	5	3.617888
5	EWR	2013	6	4.167217
6	EWR	2013	7	4.022527
7	EWR	2013	8	3.349452
8	EWR	2013	9	3.574639
9	EWR	2013	10	3.640309
10	EWR	2013	11	4.480718
11	EWR	2013	12	3.911640
12	JFK	2013	1	5.060265
13	JFK	2013	2	5.476453
14	JFK	2013	3	5.935677
15	JFK	2013	4	5.311958
16	JFK	2013	5	4.392293
17	JFK	2013	6	4.866446
18	JFK	2013	7	4.489068
19	JFK	2013	8	4.295648
20	JFK	2013	9	4.350242
21	JFK	2013	10	4.573231
22	JFK	2013	11	5.397103
23	JFK	2013	12	4.842085
24	LGA	2013	1	4.869840
25	LGA	2013	2	5.172713
26	LGA	2013	3	5.672411
27	LGA	2013	4	4.867050
28	LGA	2013	5	4.155879
29	LGA	2013	6	4.405517
30	LGA	2013	7	4.181334
31	LGA	2013	8	3.734075
32	LGA	2013	9	3.934073
33	LGA	2013	10	4.558804
34	LGA	2013	11	5.171405
35	LGA	2013	12	4.503899

Here is step:

- Identify the outlier Calculating InterQuartile
- Define the outlier threshold lower bound and upper bound
- Replace the outlier with NaN
- Group by origin, year, and month, and compute mean wind speed
- showing monthly_mean_wind_speed result

0.4.5 Q6. Draw the monthly mean wind speeds for the three airports on the same plot (three curves of different colours). Add a readable legend. Reference result

[17]:
```
# Group by origin, year, and month, and compute mean wind speed
monthly_mean_wind_speed = weather_df.groupby(['origin', 'year',␣
 ↪'month'])['wind_speed'].mean().reset_index()

#Plot line monthly wind speed
for airport in monthly_mean_wind_speed['origin'].unique():
    data = monthly_mean_wind_speed[monthly_mean_wind_speed['origin'] == airport]
    plt.plot(data['month'], data['wind_speed'], marker='o', label=airport)

# x and y label
plt.xlabel('Month')
plt.ylabel('Monthly average Wind Speed (m/s)')
plt.title('Monthly Mean Wind Speeds at NYC Airports in 2013')

# Format x-axis to show year and month (YYYY-MM)
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m'))

#Legent and showing grapha
plt.legend()
plt.tight_layout()
plt.show()
```



Step

- find the monthly_mean_wind_speed
- plot line of monthly mean wind speed Iterating the origal Airport
- adding the lable
- Format x-axis to show year and month (YYYY-MM)
- Legent and showing grapha
-

[ ]:

[ ]: